A Cumulative Damage Shock Model with Imperfect Preventive Maintenance

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In this article we consider a cumulative damage shock model under a periodic preventive maintenance (PM) policy. The PM is imperfect in the sense that each PM reduces the damage level by $100(1 - b)\%$, $0 \leq b \leq 1$. A system suffers damage due to shocks and fails when the damage level exceeds some threshold. We derive a sufficient condition for the time to failure to have an IFR distribution. We also discuss the associated problem of finding the number of PM's that minimizes the expected cost rate.

0. INTRODUCTION

In many applications, the time to failure of a system can be represented as a first passage time to a threshold for an appropriate stochastic process that describes the level of damage. There are many articles treating various models suitable for the above mechanism of failure (see [15] for a survey). Consider a system which is subject to shocks occurring randomly in time. At time $t = 0$, the damage level of the system is assumed to be 0. Upon occurrence of a shock, the system suffers a non-negative random damage. Each damage, at the time of its occurrence, adds to the current damage level of the system and, between shocks, the damage level stays constant. The system fails when the accumulated damage first exceeds a level ξ.

To keep the system in a good condition, some maintenance is periodically performed. When such maintenance takes place, it is plausible to assume that a certain amount of the damage level is reduced; otherwise, we gain no benefits from performing maintenance, which usually incurs a cost. Imperfect preventive maintenance (PM) is the notion that maintenance actions do not make a system like new but younger. Many types of imperfect PM have been considered in the literature (see, e.g., [4, 11, 13]). Recently, Kijima and Nakagawa [10] introduced a periodic imperfect PM policy where each PM reduces $100(1 - b)\%$, $0 \leq b \leq 1$, of the total damage. This article considers a cumulative-damage shock model

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described above under this PM policy. Note that if $b = 1$ then the PM is no repair at all, while if $b = 0$ then the PM coincides with a complete repair (see Kijima [8] for a related model).

When considering a cumulative damage model, one is interested in the distribution of the time to failure. A major interest in this article is also set in this direction. It should be noted that many distribution results of existing models in the literature are due to the fact that the cumulative damage process is simply a pure jump process (see, e.g., [3, 16]). In contrast to such models, the damage process of our model decreases periodically when PM actions take place. Thus, some special treatment should be required. Marshall and Shaked [12] considered a model in which, between shocks, the damage level decreases in some deterministic fashion (although the damage level never becomes negative). According to their article, the model is considered to take into account the case that the damage level may decrease between shocks due to repairs or self-recovery procedures. For this purpose, however, the model in this article seems more suitable, since the damage can be often reduced by some maintenance actions only. The aim of this article is to develop such a model and study the associated replacement problem.

The construction of this article is as follows. In the next section, we describe the model considered explicitly and give some notation. Sections 2 and 3 are devoted to the IFR property of the distribution function of the time to failure. A sufficient condition for the IFR property is given in Section 2. In Section 3, the results are applied for a Poisson shock model. Finally, in Section 4, we consider the associated replacement problem and obtain an optimal policy. For practical use, an approximate formula for the optimal policy is also given.

1. MODEL DESCRIPTION

Let $\{N(t), t \geq 0\}$ denote the nonhomogeneous Poisson process with the intensity function $\lambda(t)$ and let $\{A_n\}_{n=1}^{\infty}$ be a sequence of identically and independently distributed (iid) random damages with cumulative distribution function (CDF) $A(x)$. It is assumed that $\{A_n\}_{n=1}^{\infty}$ and $\{N(t), t \geq 0\}$ are mutually independent. Define the partial sum process $\{S_n, n \geq 0\}$ by $S_0 \equiv 0$ and $S_n = \Sigma_{k=1}^{n} A_k$. The process $\{S_{N(t)}, t \geq 0\}$ is usually called the damage process.

Suppose that PM is scheduled at periodic times kx, $k = 1, 2, \ldots$. Note that, by choosing the time scale appropriately, we can assume $x = 1$ without loss of generality. The time interval $(k - 1, k]$ will be called the kth period. We assume that PM is performed at the beginning of each period and the time needed to do it is negligible. Let D_k denote the amount of damage incurred during the kth period. It is easy to see that

$$D_k = S_{N(k)} - S_{N(k-1)}, \qquad k = 1, 2, \ldots, \tag{1}$$

and $\{D_k\}_{k=1}^{\infty}$ is a sequence of independent random variables. Note that D_k are not identical since $N(t)$ is not time homogeneous. Let Y_k denote the cumulative damage at the end of the kth period. The PM mechanism introduced previously is then represented as

$$Y_k = bY_{k-1} + D_k, \tag{2}$$

since the damage level at the beginning of period k is reduced by PM to bY_{k-1}, which is added by the total damage D_k incurred during the period. In the following, we further assume that system failure is detected only by PM. Accordingly, the time to failure is the number of PMs until a system failure is first detected.

To study the above stochastic model, we consider an absorbing Markov process that describes the behavior of the total damage since a new system is installed. Let ∞ denote an absorbing state representing a state of failure, and let $\{X_k, k \geq 0\}$ be such a process with the state space $[0,\xi] \cup \{\infty\}$ defined by $X_0 \equiv 0$ and

$$X_k = \begin{cases} bX_{k-1} + D_k, & \text{if } bX_{k-1} + D_k \leq \xi \\ \infty, & \text{otherwise} \end{cases} \tag{3}$$

[see (2)]. Let $T = \min\{k\colon X_k = \infty\}$ if the event $\{X_k = \infty\}$ is not empty and $T = \infty$ otherwise. Hence, T is the time to failure of the system. Define

$$q_k = \Pr[T = k], \quad k = 1, 2, \ldots, \infty, \quad \text{and} \quad Q_k = \sum_{n \geq k+1} q_n, \qquad k = 0, 1, \ldots, \tag{4}$$

where the sum is assumed to include ∞. Note that $\Sigma_{k\geq 1}\, q_k = 1$ so that $Q_0 = 1$. Of interest is the distribution of T. We will address this problem in Sections 2 and 3.

Now, consider the associated replacement problem. Suppose that a planned replacement is set at the Nth PM. The system is replaced by a new identical one if a failure is detected or at the Nth PM, whichever occurs first. The time interval between the installment of a system and its replacement is called a *cycle*. Let c_1 be the cost of replacement of a failed system, c_2 be the cost of planned replacement, and c_3 be the cost of PM. Since a system failure is detected only by PM, so that the duration between the actual failure and its detection may cause a loss, it is natural to assume that $c_1 > c_2$. Also, we assume $c_2 > c_3$, since otherwise no benefits of PM are obtained. Let C_N be the total cost incurred during a cycle and let $T_N = \min\{T,N\}$. It is easy to see that

$$E[T_N] = \sum_{k=1}^{N} kq_k + N \sum_{k \geq N+1} q_k = \sum_{k=0}^{N-1} Q_k \tag{5}$$

and

$$\begin{aligned} E[C_N] &= \sum_{k=1}^{N} [c_3(k-1) + c_1]q_k + \sum_{k\geq N+1} [c_3(N-1) + c_2]q_k \\ &= c_3 \sum_{k=0}^{N-1} Q_k - (c_1 - c_2)Q_N + (c_1 - c_3). \end{aligned} \tag{6}$$

The expected cost rate $L(N)$ is then given, using (5) and (6), by

$$L(N) = c_3 + \frac{c_1 - c_3 - (c_1 - c_2)Q_N}{\sum_{k=0}^{N-1} Q_k}. \tag{7}$$

If $N = 1$, i.e., the system is replaced each period, then PM is never performed and

$$L(1) = c_1(1 - Q_1) + c_2Q_1 = c_1\Pr[D_1 > \xi] + c_2\Pr[D_1 \le \xi],$$

since $Q_1 = \Pr[T \ge 2] = \Pr[D_1 \le \xi]$. The objective here is to find an optimal N^* minimizing (7). This problem will be discussed in Section 4.

2. IFR PROPERTY OF THE TIME TO FAILURE

In this section, we study the distribution of T in terms of that of D_k. For this purpose, we make use of the concept of PF_2 functions. Our discussion of the distribution of D_k in terms of $A(x)$ and $\lambda(t)$ is deferred to the next section.

Let $G_k(x) = \Pr[D_k \le x]$, $x \ge 0$, $k = 1, 2, \ldots$, and let $F_k(x) = \Pr[X_k \le x]$, $0 \le x \le \xi$, $k = 0, 1, \ldots$. $F_k(\xi)$ may be strictly less than 1 since the process $\{X_k\}$ is an absorbing process. Also, since $X_0 \equiv 0$, one has $F_0(x) = U(x)$ where $U(x) = 1$ for $x \ge 0$ and $U(x) = 0$ for $x < 0$. For convenience, we set $G_k(x) = F_k(x) = 0$ for $x < 0$ and $F_k(x) = F_k(\xi)$ for $x > \xi$. From (3), it follows that

$$F_k(x) = \int_{0-}^{\xi} G_k(x - by)\, dF_{k-1}(y), \qquad 0 \le x \le \xi. \tag{8}$$

Note that the event $\{X_k \le \xi\}$ is equivalent to the event $\{T > k\}$. It follows from (4) that

$$Q_k = F_k(\xi); \qquad q_k = F_{k-1}(\xi) - F_k(\xi), \qquad k = 1, 2, \ldots. \tag{9}$$

We will make use of the following commonly used definitions (see, e.g., [2, 5–7, 9]).

DEFINITION 2.1: A non-negative sequence $\{a_n\}_{n=-\infty}^{\infty}$ is called PF_2 in n if $a_{n_1-m_1}a_{n_2-m_2} \ge a_{n_1-m_2}a_{n_2-m_1}$ for all $n_1 < n_2$ and $m_1 < m_2$. Similarly, a non-negative function $a(x)$ is called PF_2 in x if $a(x_1 - y_1)a(x_2 - y_2) \ge a(x_1 - y_2)a(x_2 - y_1)$ for all $x_1 < x_2$ and $y_1 < y_2$.

DEFINITION 2.2: A continuous random variable X with CDF $F(x)$ is said to be IFR if the survival function $\overline{F}(x) = 1 - F(x)$ is PF_2 in x. Similarly, a discrete random variable X having $p_n = \Pr[X = n]$, $n = 1, 2, \ldots$, is called IFR if $\sum_{k=n}^{\infty} p_k$ is PF_2 in n.

Some direct implications of Definition 2.1 and known results needed for the sequel are summarized next.

PROPOSITION 2.1: (i) If a_n is either monotone or nonzero, then a local property that $a_n^2 \geq a_{n-1}a_{n+1}$ for all n is enough to ensure the PF_2 property. Also, $a(x)$ is PF_2 in x iff $a(x + y)/a(x)$ is decreasing in x for all $y \geq 0$. (Throughout this article, we use the terms "increasing" and "decreasing" in weak sense unless stated otherwise. Also, we use the convention $0/0 = 0$.) Hence, if $a(x)$ is twice differentiable, $a(x)$ is PF_2 iff $a''(x)a(x) \leq \{a'(x)\}^2$ for all x.

(ii) If a CDF is PF_2, then it is absolutely continuous except possibly at the left-hand endpoint of the interval of support (see Kijima [9]). Hence, for CDF $F(x)$ defined on $[0,\infty)$, there exists a constant c, $0 \leq c \leq 1$, and a probability density function (PDF) $f(x)$ such that $F(x) = c + (1 - c)\int_0^x f(t)\,dt$. A sufficient condition for $F(x)$ to be PF_2 is that $f(x)$ is decreasing in x.

(iii) The survival function being PF_2 does not imply that its CDF is PF_2, and vice versa. If the CDF has a PDF which is PF_2, then both the CDF and the survival function are necessarily PF_2 [5].

REMARK 2.1: (i) From (1), it is readily seen that each D_k has a probability mass at 0. Hence the class of PF_2 distributions is suitable for a Poisson shock model (see Section 3).

(ii) The density functions of generalized Erlang, normal and folded normal distributions, and Weibull distributions with shape parameter not less than 1 are PF_2. Thus, the distribution functions of these distributions are PF_2.

(iii) A continuous random variable X with CDF $F(x)$ is said to be DFR if $\bar{F}(x + y)/\bar{F}(x)$ increases in x for all $y \geq 0$. It is known (see, e.g., [1]) that $F(x)$ is absolutely continuous except possibly at the left-hand endpoint of the interval of support. Hence, as in Proposition 2.1(ii), $F(x)$ can be written as $\bar{F}(x) = (1 - c)\int_x^\infty f(t)\,dt$. It then follows from the definition that $f(x)$ is decreasing in x and the support of f is $[0,\infty)$ (see Barlow and Proschan [1]). Hence, from Proposition 2.1(ii), any DFR distribution is PF_2. Moreover, $f(x)$ is strictly decreasing, since $f'(x)\bar{F}(x) \leq -\{f(x)\}^2 < 0$, if $f'(x)$ exists.

DEFINITION 2.3: Let f and g be non-negative functions (sequences, respectively). f is said to be uniformly smaller than g, which we denote $f < g$, if

$$f(x)g(y) \geq f(y)g(x)\ (f_n g_m \geq f_m g_n), \qquad \text{for all } x < y\ (n < m). \tag{10}$$

REMARK 2.2: For two distribution functions $F(x)$ and $G(x)$, the concept of $F < G$ was first introduced by Keilson and Sumita [6] (they called $<$ uniform ordering in the negative direction). It is easily seen that $<$ defines a partial ordering. Moreover, $F < G$ implies that $F(x) \geq G(x)$ for all x, i.e., F is stochastically smaller than G. The ordering $<$ is now frequently used (see, e.g., [6, 7, 9, 14]).

Consider the integral transformation

$$(TF)(x) = \int_{-\infty}^{\infty} P(y,x)\,dF(y) \tag{11}$$

for an increasing function $F(x)$ defined on $(-\infty,\infty)$. If $P(y,x)$ is TP_2 in y and x, i.e. $P(y_1,x_1)P(y_2,x_2) \geq P(y_1,x_2)P(y_2,x_1)$ for all $y_1 < y_2$ and $x_1 < x_2$, then the kernel $P(y,x)$ is called uniformly monotone [9]. The next lemma is due to Kijima [9], but can be shown by mimicking the proof of Lemma 2.3 in Shaked and Shanthikumar [16].

LEMMA 2.1: Let $F(x)$ and $G(x)$ be non-negative and increasing functions such that $F < G$. Then $P(y,x)$ is uniformly monotone iff $TF < TG$.

The next corollary is immediate.

COROLLARY 2.1: Let $F(x)$, $G(x)$, and $H(x)$ be CDFs. Suppose $H(x)$ is PF_2 in x and $F < G$. Then, $H * F < H * G$, where $*$ denotes the convolution; i.e., $H * F(x) = \int_{-\infty}^{\infty} H(x - y)\, dF(y)$.

In the next theorem, we provide a sufficient condition on D_k for which T is IFR.

THEOREM 2.1: Let $G_k(x)$ be the CDF of D_k, $k = 1, 2, \ldots$. Suppose $G_1 < G_2 < \cdots$ and $G_k(x)$ are PF_2 in x. Then Q_k in (9) is PF_2 in k for any $\xi > 0$. Thus, T is IFR.

PROOF: Let $H_k(x) = F_k(x/b)$ and consider

$$\hat{F}_k(x) = \int_{-\infty}^{\infty} G_k(x - by)\, dF_{k-1}(y)$$

$$= \int_{-\infty}^{\infty} G_k(x - y)\, dH_{k-1}(y), \qquad -\infty < x < \infty. \tag{12}$$

Then $\hat{F}_k(x)$ is increasing in x and, from the definition of $F_k(x)$, $-\infty < x < \infty$, one sees that $F_k(x) = \hat{F}_k(x)$ for $x \le \xi$ and $F_k(x) = \hat{F}_k(\xi)$ for $x > \xi$. Also, it is not hard to see that if $\hat{F}_k(x)$ is PF_2 in x, then so is $F_k(x)$. Note that $F_0(x) = U(x)$ is PF_2 in x. Now suppose that $F_{k-1}(x)$ is PF_2 in x. Evidently, $H_{k-1}(x)$ is also PF_2 in x. This is so iff $H_{k-1}(\cdot - x_1) < H_{k-1}(\cdot - x_2)$ for all $x_1 < x_2$. From Corollary 2.1, it follows that $\hat{F}_k(\cdot - x_1) < \hat{F}_k(\cdot - x_2)$, since $G_k(x)$ in PF_2 in x. Hence, $\hat{F}_k(x)$ is PF_2 in x, and so is $F_k(x)$. Thus, $F_k(x)$ is PF_2 in x for all k by induction. Next, suppose $F_{k-1} < F_k$. Then $H_{k-1} < H_k$. Certainly, this holds for $k = 1$. From Corollary 2.1 and the assumptions, it then follows that

$$\hat{F}_k < \int_{-\infty}^{\infty} G_k(\cdot - y)\, dH_k(y) = \int_{-\infty}^{\infty} H_k(\cdot - y)\, dG_k(y)$$

$$< \int_{-\infty}^{\infty} H_k(\cdot - y)\, dG_{k+1}(y) = \hat{F}_{k+1},$$

from which $F_k < F_{k+1}$. Therefore, $F_0 < F_1 < \cdots$.

Consider now (8), with $x = \xi$. Then, from (9) one sees that

$$\frac{Q_k}{Q_{k-1}} = \int_{0-}^{\xi} \frac{F_{k-1}[(\xi - y)/b]}{F_{k-1}(\xi)}\, dG_k(y). \tag{13}$$

For y such that $(\xi - y)/b > \xi$, one has

$$\frac{F_{k-1}[(\xi - y)/b]}{F_{k-1}(\xi)} = \frac{F_k[(\xi - y)/b]}{F_k(\xi)} = 1, \tag{14}$$

by definition. For y such that $(\xi - y)/b \le \xi$,

$$\frac{F_{k-1}[(\xi - y)/b]}{F_{k-1}(\xi)} \ge \frac{F_k[(\xi - y)/b]}{F_k(\xi)}, \tag{15}$$

since $F_{k-1} < F_k$. Thus, combining (13)–(15) yields

$$\begin{aligned} \frac{Q_k}{Q_{k-1}} &\ge \int_{0-}^{\xi} \frac{F_k[(\xi - y)/b]}{F_k(\xi)}\, dG_k(y) \\ &\ge \int_{0-}^{\xi} \frac{F_k[(\xi - y)/b]}{F_k(\xi)}\, dG_{k+1}(y) = \frac{Q_{k+1}}{Q_k}, \end{aligned} \tag{16}$$

where the second inequality in (16) follows since $G_k < G_{k+1}$ implies $G_k(x) \ge G_{k+1}(x)$ for all x. Hence, the result follows by Proposition 2.1(i). □

3. POISSON SHOCK MODEL

In Section 2, we obtained the IFR property of T under specific assumptions imposed on D_k. In order to apply this result to the Poisson shock model described in Section 1, we need to obtain conditions on $A(x)$ and $\lambda(t)$ so that D_k satisfy the conditions in Theorem 2.1.

Let N follow a Poisson distribution with the mean λ and let $\{A_n\}_{n=1}^{\infty}$ be a sequence of iid random damages as in Section 1. Consider the compound Poisson distribution of $\Sigma_{n=1}^{N} A_n$, where $\Sigma_{n=1}^{0} A_n \equiv 0$. We denote its CDF by $G_\lambda(x)$, where

$$G_\lambda(x) = \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} e^{-\lambda} A^{(n)}(x), \qquad x \ge 0. \tag{17}$$

Here $A^{(n)}(x)$ denotes the n-fold convolution of $A(x)$ with itself. $G_\lambda(x)$ has a positive mass at $x = 0$. Also, $G_\lambda(x)$ is infinitely divisible; i.e., $G_\lambda(x) = G_{\lambda/n}^{(n)}(x)$ for any $n \ge 1$. Note that if λ is the mean number of shocks in the kth period then $G_\lambda(x)$ coincides with the CDF $G_k(x)$ of D_k.

We first obtain a sufficient condition for $G_k(x)$, $k = 1, 2, \ldots,$ to be PF_2 in x.

LEMMA 3.1: For some c, $0 \leq c < 1$, and a PDF $a(x)$, suppose $A(x) = c + (1 - c) \int_0^x a(t)\, dt$. If $a(x)$ is strictly decreasing in x on $[0,\infty)$ then $G_\lambda(x)$ in (17) is PF_2 in x.

PROOF: For sufficiently small $\lambda > 0$, one has

$$\Pr[N = 0] = e^{-\lambda}, \qquad \Pr[N = 1] = \lambda e^{-\lambda}, \qquad \text{and} \quad \Pr[N \geq 2] = o(\lambda), \tag{18}$$

where $o(\lambda)$ denotes the small order of λ. It then follows from (17) that

$$G_\lambda(x) = e^{-\lambda} + \lambda e^{-\lambda} A(x) + o(\lambda), \qquad x \geq 0. \tag{19}$$

For $G_\lambda(x)$ to be PF_2 in x, it is sufficient that

$$\frac{1 + \lambda A(x + y) + o(\lambda)}{1 + \lambda A(x) + o(\lambda)}$$

strictly decreases in x for any $y \geq 0$ [see Proposition 2.1(ii)]. A sufficient condition for that is

$$A(x + y) - A(x) \quad \text{strictly decreases in } x \text{ for any } y \geq 0. \tag{20}$$

Hence, if the PDF $a(x)$ is strictly decreasing in x, (20) holds, and thus $G_\lambda(x)$ is PF_2 in x for sufficiently small λ. Now, for any $\lambda > 0$, there exists n_0 such that $G_{\lambda/n}(x)$ is PF_2 in x for any $n \geq n_0$. It follows from Theorem 5.3 in Chapter 3 of Karlin [5] that $G_\lambda(x) = G_{\lambda/n}^{(n)}(x)$ is PF_2 in x. Hence, the lemma is proved. □

Note added in the proof: There is a flaw in the proof of Lemma 3.1, which affects Remark 3.1(i) and Theorem 3.1. For Theorem 3.1 to be valid, it seems that we must look for other conditions for the compound Poisson distribution to be PF_2. This problem is open. Theorem 4.1 holds as far as T is IFR.

REMARK 3.1: (i) From Remark 2.1(iii), if the random damages follow a DFR distribution, then the associated compound Poisson distribution is PF_2. Hence, exponential random damages satisfy the condition.

(ii) The condition in Lemma 3.1 is a sufficient condition. Indeed, it is easily seen that when $A_n \equiv 1$; i.e., each shock causes a constant damage, $G_\lambda(x)$ is PF_2 in x. Any closure property of distribution classes seems not known for compound Poisson distributions.

Suppose that the intensity function $\lambda(t)$ of the nonhomogeneous Poisson process in Section 1 is increasing. This represents the realistic situation where the older the system the more frequently shocks occur. Under this assumption, one can show that $G_1 < G_2 < \cdots$.

LEMMA 3.2: Suppose that $\lambda(t)$ is increasing in t and $A(x)$ is PF_2 in x. Then, $G_1 < G_2 < \cdots$, where $G_k(x)$ are the CDFs of D_k.

PROOF: Let $\Lambda(k) = \int_{k-1}^{k} \lambda(t)\, dt$, $k = 1, 2, \cdots$. Since $\lambda(t)$ increases in t, $\Lambda(1) \leq \Lambda(2) \leq \cdots$. Let M and N follow the Poisson distributions with the means $\Lambda(k)$ and $\Lambda(k + 1)$, respectively. Define $P_n^M = \Pr[M \leq n]$ and $P_n^N = \Pr[N \leq n]$, $n = 0, 1, \ldots$, with $P_{-1}^M = P_{-1}^N = 0$. Then, it is easy to see that $P_n^M P_{n+1}^N \geq P_{n+1}^M P_n^N$; i.e., $P^M < P^N$. Note that $D_k \overset{d}{=} \Sigma_{n=1}^M A_n$ ($\overset{d}{=}$ stands for equality in law), and hence

$$G_k(x) = \sum_{n=1}^{\infty} A^{(n)}(x)\, d[P_n^M] = (TM)(x),$$

where $d[P_n^M] = P_n^M - P_{n-1}^M$. Hence, applying Lemma 2.1, one sees that $TM < TN$ since $A^{(n)}(x)$ is TP_2 in n and x (see, e.g., [5, 9]). Thus, $G_k < G_{k+1}$, proving the lemma. □

Combining Theorem 2.1 and Lemmas 3.1 and 3.2, one has the next main theorem.

THEOREM 3.1: In the Poisson shock model described in Section 1, if $\lambda(t)$ is increasing in t and if $A(x)$ has a strictly decreasing PDF on $[0,\infty)$, then the time to failure T has the IFR property.

4. OPTIMAL POLICY AND APPROXIMATION

We have seen that, under certain conditions, the time to failure T has the IFR property. This property is important in the replacement problem since it guarantees the uniqueness of the optimal policy, as we shall see later. In this section, we discuss an optimal policy for the replacement problem described in Section 1.

Forming the inequalities $L(N) \leq L(N + 1)$ and $L(N) < L(N - 1)$ in (7) one has

$$l(N) \geq C \quad \text{and} \quad l(N - 1) \leq C, \tag{21}$$

where $C = (c_1 - c_3)/(c_1 - c_2) > 1$, $l(0) \equiv 1$ and

$$l(N) = \sum_{k=0}^{N} Q_k - \frac{Q_{N+1}}{Q_N} \sum_{k=0}^{N-1} Q_k, \qquad N \geq 1. \tag{22}$$

Hence, if $l(N)$ is increasing in N, then a solution satisfying (21) can be uniquely determined by choosing the minimum. In order for $l(N)$ to be increasing, the PF_2 property of Q_k in k is required, since

$$l(N + 1) - l(N) = \left(\frac{Q_{N+1}}{Q_N} - \frac{Q_{N+2}}{Q_{N+1}}\right) \sum_{k=0}^{N} Q_k \geq 0$$

iff Q_k is PF_2 in k. Thus, the conditions of Theorem 3.1 guarantee the monotonicity of $l(N)$. In this section, we assume that the conditions of Theorem 3.1 hold.

Let $m = E[T] = \Sigma_{k=0}^{\infty} Q_k$ and suppose that there exists a finite k such that $Q_k < 1$. Let k^* be the minimum of k's such that $Q_k < 1$ and write $\rho = Q_{k^*}$. Then,

$$Q_{N+k^*} = \frac{Q_{N+k^*}}{Q_{N+k^*-1}} \cdots \frac{Q_{k^*+1}}{Q_{k^*}} Q_{k^*} \leq \rho^{N+1}, \qquad N \geq 0,$$

since Q_{k+1}/Q_k is decreasing in k. Therefore, unless $A_n \equiv 0$ almost surely for all k; i.e., damages never occur, it is assured that Q_k approaches to 0 geometrically fast as $k \to \infty$. Hence $m < \infty$. It then follows from (22) that

$$l(\infty) \equiv \lim_{N\to\infty} l(N) = m \lim_{N\to\infty} \left(1 - \frac{Q_{N+1}}{Q_N}\right). \tag{23}$$

Hence, if $l(\infty) > C$ or, equivalently, $\lim_{N\to\infty} F_{N+1}(\xi)/F_N(\xi) < 1 - m^{-1}C$ then there exists a finite and unique N^* satisfying (21). On the other hand, if $l(\infty) \leq C$ then $N^* = \infty$. Thus, the next theorem follows.

THEOREM 4.1: Under the conditions of Theorem 3.1:
(i) If $\lim_{N\to\infty} F_{N+1}(\xi)/F_N(\xi) < 1 - m^{-1}C$ then there exists a finite and unique N^* satisfying (21).
(ii) If $\lim_{N\to\infty} F_{N+1}(\xi)/F_N(\xi) \geq 1 - m^{-1}C$ then $N^* = \infty$; i.e., the system should be replaced only at failure. Thus, if $C \geq m$ then $N^* = \infty$.

It is, in general, not easy to calculate $Q_k = F_k(\xi)$ numerically for all k via (8). Accumulated error from numerical integration may become intolerable especially for large k. Accordingly, it seems very difficult to obtain optimal N^* based on (21) and (22) if N^* is very large. Therefore, need for an approximation of N^*, which is easy to evaluate, arises. In the following, we assume that $\lambda(t) = \lambda$ for simplicity. Accordingly, D_k are iid with CDF $G(x)$. The mean $m = E[T]$ is then evaluated by solving the integral equation

$$m(x) = 1 + \int_{0-}^{\xi-x} m(bx + by)\, dG(y), \qquad 0 \leq x \leq \xi. \tag{24}$$

Namely, $m = m(0)$. Using this m, we develop a simple approximation formula of N^*.

Let $r_k = q_k/Q_k$. Since Q_k is PF_2 in k, r_k is increasing in k. Also, $\log Q_{k-1} - \log Q_k = \log(1 + r_k)$, resulting in $-\log Q_k = \Sigma_{n=1}^{k} \log(1 + r_n)$. Hence $-(1/k) \log Q_k$ is increasing in k (Barlow and Proschan [2] called a continuous function f star-shaped if $-(1/t) \log f(t)$ is increasing in t). It follows from similar arguments employed in [2] that, for any $0 < \alpha < 1$, Q_k crosses α^k at most once, and if it crosses, it does so from above. Now choose $\alpha = 1 - m^{-1}$. Then, $\Sigma_{k=0}^{\infty} \alpha^k = m = \Sigma_{k=0}^{\infty} Q_k$. Thus, $\Sigma_{k=0}^{N} Q_k \geq \Sigma_{k=0}^{N} (1 - m^{-1})^k$ for all $N \geq 0$. On the other hand, from (22), $l(N)$ can be rewritten as

$$l(N) = \left(1 - \frac{Q_{N+1}}{Q_N}\right) \sum_{k=0}^{N} Q_k + Q_{N+1}. \tag{25}$$

We define

$$\hat{l}(N) = \sum_{k=0}^{N} (1 - m^{-1})^k = m[1 - (1 - m^{-1})^{N+1}]. \tag{26}$$

Note that $1 \geq (1 - Q_{N+1}/Q_N)$ and $\Sigma_{k=0}^{N} (1 - m^{-1})^k$ underestimates $\Sigma_{k=0}^{N} Q_k$. Q_{N+1} is negligible for large N. Hence, comparing (25) and (26), $\hat{l}(N)$ may be close to $l(N)$. Therefore, we have an approximation value $\hat{N}$of N^* that is obtained as the smallest N such that $\hat{l}(N) \geq C$; i.e.,

$$\hat{N} = \inf\left\{N\colon N \geq \frac{\log(1 - m^{-1}C)}{\log(1 - m^{-1})} - 1\right\}. \tag{27}$$

It should be noted that, if $C \geq m$, then $\hat{N} = \infty$ (see Theorem 4.1).

5. CONCLUDING REMARKS

In this article, we develop a new cumulative damage shock model under a periodic PM policy. A system suffers damage due to shocks and fails when the damage level exceeds a threshold. In contrast to existing models, the damage process is not merely additive since each PM reduces $100(1 - b)\%$ of the total damage. This new formulation makes the damage process up and down so as to introduce some difficulties in analyzing the associated stochastic process. The notion of uniform monotonicity is invoked for this purpose. A sufficient condition under which the time to failure has an IFR property is obtained. The IFR property is important in discussing the associated replacement problem since it guarantees the uniqueness of the optimal policy. Finally, we show how to compute the optimal policy for the replacement problem and give a method for approximating the optimal number of PMs.

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